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30 APRIL 1965
SUPERSEDING
FA-PD-MI-2224, Rev. 1
8 MAY 1959

MILITARY SPECIFICATION

CARTRIDGE, INITIATOR, M69

1. SCOPE

1.1 This specification covers Cartridge, Initiator, M69, for use in conjunction with other propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of the specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-A-2550 — Ammunition and Special Weapons, General Specification for.

MIL-E-5272 — Environmental Testing, Aeronautical and Associated Equipment, General Specification for.

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-109 — Quality Assurance Terms and Definitions.

MIL-STD-639 — Surface Roughness Values Preferred for Army Ordnance Materiel.

MS-28741 — Hose Assembly, Detachable End-Fitting, Medium Pressure.

DRAWINGS

U. S. ARMY MUNITIONS COMMAND

D8593496 — Cartridge, Initiator, M69, Assembly.

F8797469 — Box, Packing, Ammunition, for Cartridge, Initiator, M69.

F8797470 — Carton, Packing, Ammunition, for Cartridge, Initiator, M69.

IEL-8593496 — Index of Inspection Equipment List for Cartridge, Initiator, M69.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained

FSC 1375

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from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General. Materials, parts and assemblies shall conform to the applicable drawings, referenced specifications, and the following:

3.2 Initial production sample. Unless otherwise directed by the contracting officer, an initial production sample shall be required.

3.3 Protective finish. Protective finish shall be in accordance with the applicable specifications and drawings and shall apply to all surfaces of the component regardless of configuration.

3.4 Vibration. The assembled cartridge shall withstand vibration as prescribed in MIL–E–5272, Procedure XII, without evidence of any disintegration of the primer, igniter or propellant or derangement of components.

3.5 Functional requirements.

3.5.1 The cartridge shall not fail to fire.

3.5.2 *Powder charges.* The igniter and propellant charges shall be fixed by the contractor to produce the specified ballistic requirements (see 6.1).

3.5.3 *Ballistics.* The cartridge when tested in an M9 Initiator and temperature conditioned in accordance with Table I shall produce the pressures specified therein at the end of 30 feet of aircraft hose, size No. 4, MS–28741, in a one cubic inch pressure station.

TABLE I

Temperature degrees Fahrenheit (°F.)	Pressure pounds per square inch (PSI)		Ignition delay maximum (see 6.2)
	Minimum	Maximum	
−65° to −70°	1000	2200	65
65° to 75°	1200	2300	65
160° to 165°	1300	2400	65

3.6 Workmanship. Workmanship shall be in accordance with the requirements of the applicable drawings, referenced specifications and the following:

3.6.1 *Metal defects.* The metal parts assembly shall be free from cracks, splits, cold shuts, inclusions, porosities or other metal defects.

3.6.2 *Bur.* The part or assembly shall not have a bur which might interfere with assembly or function of the item or which might be injurious to personnel handling the cartridge.

3.6.3 *Foreign matter.* The part or assembly shall not contain chips, dirt, grease, oil, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 *Inspection terms and definitions.* Reference shall be made to MIL–STD–109 to define the quality assurance terms used.

4.1.2 *Inspection.* Inspection shall be in accordance with MIL–A–2550.

4.2 Initial production sample. At the beginning of regular production, the contractor shall submit an initial production sample to a Government approved facility for evaluation. The sample shall consist of 30 cartridges and 10 sets of unassembled components including explosive components. The sample shall be manufactured in the same manner, using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing must be the same as used in regular production and shall be obtained from the same source of supply. The contractor shall inspect the sample in accordance with his inspection system. After inspection and provisional acceptance at source, the initial production sample shall be inspected at the Government approved facility for all requirements of the drawings and specifications. The Government reserves the right to require new initial production samples until such time as an acceptable sample is submitted.

4.3 Inspection provisions.

4.3.1 *Lot.*

4.3.1.1 *Submission of product.* The product shall be submitted in accordance with MIL–STD–105.

4.3.1.2 *Size of lot.* Unless otherwise specified by the procuring agency an ammunition lot shall consist of 600 cartridges, maximum, exclusive of test samples, and shall consist of:

(a) Components produced by a homogeneous process.

(b) Propellant from not more than one lot.

(c) Igniter from not more than one lot.

(d) Primers from not more than one lot.

4.3.2 *Examination.* One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects, using the sampling plans and criteria in tables II and III. For an individual defect (single characteristic), the criteria in Table II shall be used. For total defects within a major or minor group, the criteria in Table III shall be used. All nonconforming material shall be rejected. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter. The Ordnance Standard Finish Blocks prescribed in MIL–STD–639 shall be used as a basis of comparison for surface roughness determination.

TABLE II

Lot size	Sample size	Acceptance Numbers	
		Major	Minor
1–300	100% Examination	0	0
301–1300	110	1	2
1301–3200	150	2	3

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TABLE III

Number of defects in class	Total defects allowable
	Major
2– 3	1
4– 7	2
8–19	3
	Minor
2– 3	2
4– 7	3
8–16	4

4.3.2.1 *Cartridge, Initiator, M69, Assembly, Drawing D8593496.*

	Defect	*Method of Inspection*
CRITICAL		
1.	Any part missing	X-Ray and Visual [1]
2.	Improper assembly	X-Ray and Visual [1]
3.	Mixed ammunition	Visual
4.	Metal defective	Visual
5.	Crimp missing or incorrect	Visual
6.	Marking missing, incorrect or illegible	Visual
MAJOR		
101.	Cartridge profile failure	Gage
102.	Diameter of body, max.	Gage
103.	Diameter of head, max.	Gage
104.	Length to datum diameter, max.	Gage
105.	"O" ring split, cracked or cut	Visual
106.	Protective finish inadequate or defective	Visual
107.	Foreign matter	Visual
108.	Burr	Visual
MINOR		
201.	Total length	Gage

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Defect	*Method of Inspection*
202. Dent	Visual
203. Draw scratch	Visual
204. Fold, wrinkle, buckle or bulge	Visual
205. Varnish inadequate or missing	Visual
206. Surface finish improper	Visual

[1] The assembled cartridge shall be X-rayed with the long axis of the cartridge perpendicular to the rays of the X-ray machine. The X-ray negatives, properly identified, shall be forwarded to the procuring agency.

4.3.2.2 *Primer, Percussion, 50M, Dwg. C7645339 (prior to assembly in cartridge).*

Defect	*Method of Inspection*
CRITICAL	
1. Improper assembly	X-ray [1]
2. Any part missing	X-ray [1]
3. Mixed primers	Visual
4. Primer pellet broken, cracked or dislodged	Visual
5. Primer foil discolored	Visual
6. Metal defective	Visual
MAJOR	
101. Burr	Visual
102. Foreign matter	Visual
MINOR	
None defined	

[1] Prior to assembly into a cartridge the primer shall be X-rayed with the sides of the cup perpendicular to the rays. The X-ray negatives, properly identified, shall be forwarded to the procuring activity.

4.3.2.3 *Carton, Packing, Ammunition, Drawing F8797470.*

Defect	*Method of Inspection*
CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Quantity of contents incorrect	Visual
202. Contents loose	Manual and Visual

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Defect	Method of Inspection
203. Separator or filler missing	Visual
204. Heat sealer incomplete, loose or distorted	Visual
205. Cut or damage through impregnated box	Visual
206. Marking missing, incorrect or illegible	Visual

4.3.2.4 *Box, Packing, Ammunition, Drawing F8797469.*

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Contents loose	Manual and Visual
202. Hardware missing, loose, broken or improperly engaged	Visual
203. Strapping missing, loose, broken or improperly assembled	Visual
204. Contents of box exposed	Visual
205. Box damaged	Visual
206. Marking missing, incorrect or illegible	Visual
207. Number of cartons incorrect	Visual
208. Car Seal missing, unsealed or improperly positioned	Visual

4.3.3 *Inspection testing.* The following tests shall be performed in accordance with the provisions of 4.4.

4.3.3.1 *Protective finish.* The number of specimens and acceptance criteria for protective finish tests shall be in accordance with the applicable military standard listed on the drawing.

4.3.3.2 *Vibration.* Twelve of the 30 cartridges of initial production sample shall be subjected to vibration. The 12 cartridges shall then be x-rayed in accordance with note 1 of 4.3.2.1. Failure of any cartridge to comply with 3.4 shall be cause for rejection of the lot. Any change in the lot of primers in subsequent production shall be reason to select an additional sample of 12 cartridges and subject it to the vibration test.

4.3.3.3 *Airtightness.* Each cartridge shall be subjected to the specified pressure differential. Failure to comply with the requirements of the drawing shall be cause for rejection of the cartridge.

4.3.3.4 *Functional.* The following test shall be performed by the Government.

4.3.3.4.1 *Ballistics.* Thirty assembled cartridges from the initial production sample and 30 cartridges from each subsequent production lot shall be functionally tested. Failure of any cartridge to comply with the requirements of 3.5.3 shall be cause for rejection of the lot.

TABLE IV. *Sample Size*

Temperature	Production Lot
−65° to −70° F	10
65° to 70° F	10
160° to 165° F	10

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4.3.3.5 *Sealant application.* Five cartridges from each production lot shall be selected for this test. Failure of any cartridge to comply with the drawing requirements shall be cause for rejection of the lot of cartridges.

4.3.4 *Inspection equipment.* The examination and tests shall be made using equipment listed on IEL–8593496.

4.4 Test methods and procedures.

4.4.1 *Protective finish.* Test procedures for protective finish shall be in accordance with the applicable military standard listed on the drawing.

4.4.2 *Airtightness.* The method for determining airtightness shall be by use of equipment listed on IEL–8593496.

4.4.3 *Vibration.* The method and procedure for performing the vibration test shall be in accordance with MIL–E–5272, Procedure XII.

4.4.4 *Functional.*

4.4.4.1 *Ballistics.* The cartridge shall be loaded into an acceptable M9 Initiator, divided into 3 equal groups and temperature conditioned in accordance with table IV. The initiator shall have end caps to cover the gas port during the conditioning period. The cartridge shall be tested within 5 minutes after removal from the conditioning chamber (see fig. 1). When this time is exceeded the initiator shall be reconditioned at the original temperature.

4.4.4.1.1 *Temperature conditioning.* A standard M9 Initiator loaded with a dummy M69 Cartridge, containing a temperature sensitive element, shall be placed in the conditioning chamber at the same time as the samples. The conditioning time shall be a minimum of 1 hour longer than the time necessary to bring the element contained in the dummy cartridge to the required temperature.

4.4.4.2 *Sealant application.* The cartridge shall be disassembled in such a manner that determination can be made that the sealant has been applied for the full length of the joining surfaces at the junction of the sealing disc and case sidewall and that the propellant cavity is free of sealant.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking. Packing and marking shall be in accordance with Drawings F8797469 and F8797470, or as required by contract.

5.2 Data cards. Data card information shall be in accordance with MIL–A–2550.

6. NOTES

6.1 Advisory propellant loading data:

Black powder, Grade A–4	15.3 grains ± 0.3 grain
Propellant, Class (M2)	134.0 grains ± 5.0 grains
Propellant Grain Dimensions:	
Length	0.484 inches
Diameter	0.250 inches
Perforation diameter	0.061 inches
Web thickness	0.094 inches

6.2 Ignition delay is defined as the time that elapses between the firing of the primer and the beginning of a continuous pressure-rise on a pressure time record.

Custodian:

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Preparing activity:

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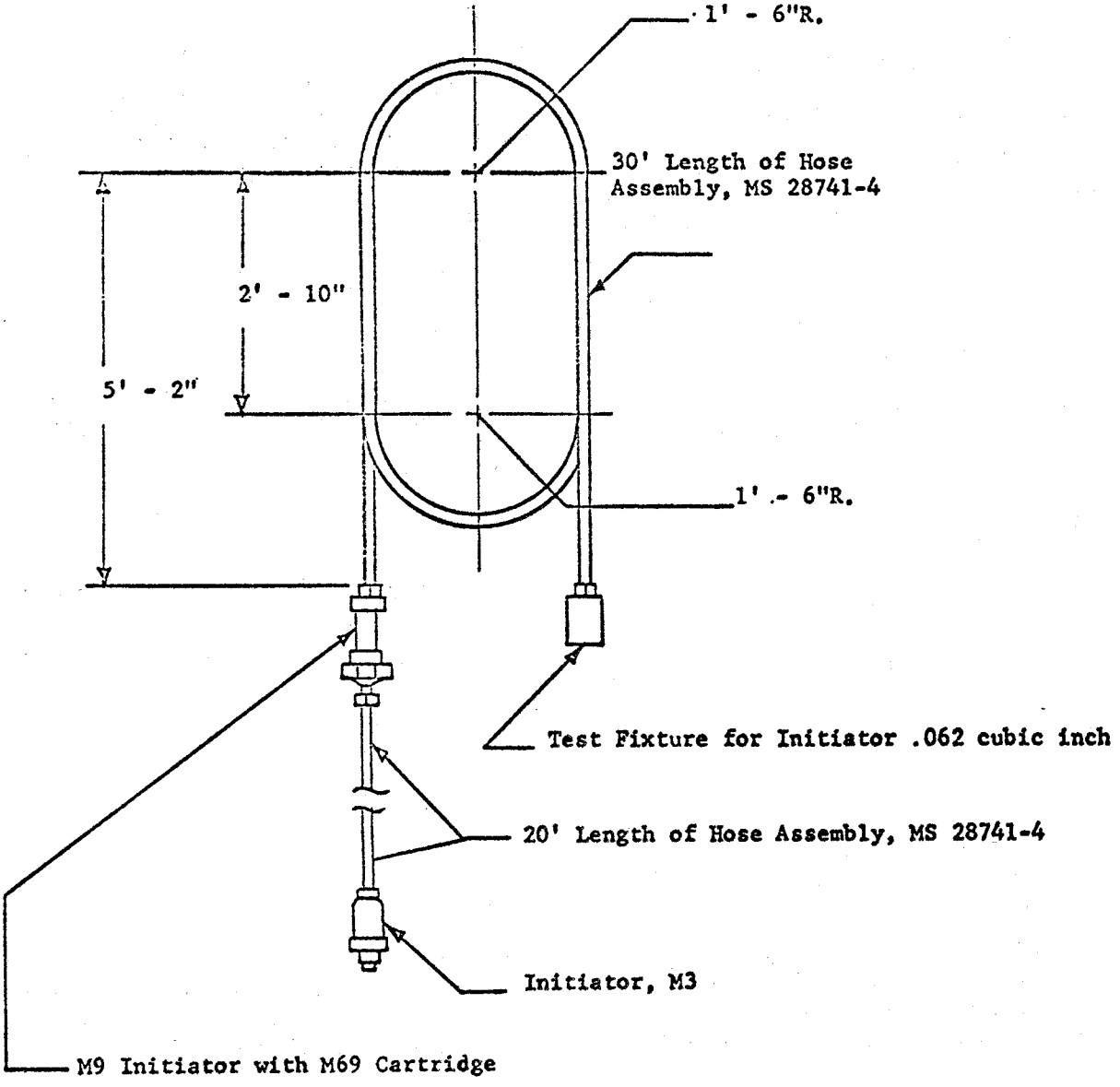


Figure 1

PERFORMANCE TEST LAYOUT SKETCH

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION: Cloth, Oxford, Cotton Warp and Nylon Filling, Quarpel Treated MIL-C-003924C(GL)

ORGANIZATION | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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